

GENERAL SPECIFICATION G7.18

Two-Way Voice Communication Systems (Fireman's Intercom)

CONTENTS

G7.18	Two-Way Voice Communication Systems (Fireman's Intercom).....	1
G7.18.1	General Requirements.....	1
G7.18.2	Facilities	1
G7.18.3	Master Control Station	1
G7.18.4	Remote Stations.....	3
G7.18.5	System Operation	4
G7.18.6	Testing & Commissioning.....	4

G7.18 TWO-WAY VOICE COMMUNICATION SYSTEMS (FIREMAN'S INTERCOM)

G7.18.1 General Requirements

- (a) Provide a two-way voice communication system that will allow reliable communication between the Fire Command Centre and each remote handset station. Provide a system that is independent of any other communications system.
- (b) Provide a system that complies fully with the requirements of SS CP 25, Code of Practice for Emergency Voice Communication System in Buildings.

G7.18.2 Facilities

- (a) Operation
 - (i) Provide a system with the following operation facilities:
 - 1) Two-way communication;
 - 2) Selective zone calling;
 - 3) Private conversation with selected remote station handsets; and
 - 4) Conference call to multiple remote station handsets.
- (b) Technical
 - (i) Provide a system with the following technical facilities:
 - 1) Automatic supervision of the complete system with audible and visual indication of internal system faults;
 - 2) Automatic changeover to battery supply in the event of mains failure; and
 - 3) Modular construction to permit future expansion.

G7.18.3 Master Control Station

- (a) General
 - (i) Provide a master control station where indicated on the drawings. House the master control station in a red-painted steel cabinet, suitable for flush or surface mounting as indicated.
 - (ii) Provide full details and drawings for approval by the SO before commencing any work or placing any orders.
 - (iii) Ensure the master control station provides as a minimum, the facilities as mentioned below:
 - 1) Master handset;
 - 2) Remote station selection;
 - 3) Batteries and charger;
 - 4) Fault monitoring; and
 - 5) Voice recorder.

- (b) Master Handset
 - (i) Provide a master handset module, fire red in colour, with audible buzzer for incoming calls, a busy/call tone generator and a push-button to permit private conversation.
- (c) Remote Station Selection
 - (i) Provide a zone selection module with pushbuttons for selection of remote stations. Provide monitoring/supervision of all interconnecting cabling. Provide individual indication for fault and call status to each remote station.
 - (ii) Provide a sufficient number of manufacturer's standard modules to cover the number of remote stations. Label all zones with a permanent indelible label. Label all unused zones accordingly.
- (d) Batteries and Charger
 - (i) Provide a combined power supply and battery charger with boost and trickle charge rates suitable for the batteries supplied. Unless indicated otherwise provide sealed lead acid batteries to BS 6290-4. Ensure the capacity of the battery is sized such that it is sufficient to operate the system for not less than 4 hours.
 - (ii) Provide a switch for manual selection of boost charge, together with automatic return to trickle charge rate should the charger be left on the boost setting for a period that may cause damage to the batteries.
 - (iii) Provide LED indicators for mains on, battery or charger fault and boost charge selected.
- (e) Fault Monitoring
 - (i) Provide an audible alarm and monitoring for the following as a minimum:
 - 1) Master system fault;
 - 2) Remote system fault;
 - 3) Battery fault; and
 - 4) Battery charger fault.
 - (ii) Provide a pushbutton to mute the audible alarm. Ensure that the audible alarm operates in the event of a second alarm condition.
- (f) Voice Recorder
 - (i) Where indicated provide a two-channel voice recorder to permit concurrent recording of all alarm signals and voice traffic over the PA system on one channel and voice traffic over the emergency intercom on the other.
 - (ii) Provide a suitable interface at the master control situation with the following facilities as a minimum:
 - 1) Indication of input from fire alarm;
 - 2) Indication of input from emergency paging system;
 - 3) Indication of input from fireman's intercom;
 - 4) Recording status;

- 5) Recording medium not present;
- 6) Recording medium full; and
- 7) Digital indication of total recording time, with reset pushbutton.

G7.18.4 Remote Stations

- (a) General
 - (i) Provide remote handset stations where shown on the drawings. Where damage or vandalism may be expected, or where specifically indicated, provide remote handset jack stations that can be used in conjunction with a portable handset.
- (b) Remote Handset Stations
 - (i) Provide flush mounted or surface mounted stations.
 - (ii) Provide remote stations with handset housed in a lockable enclosure constructed from minimum 16 gauge steel with a breakable glass front. Ensure all stations are fitted with locks to suit a master key.
 - (iii) Provide an appropriate number of master keys for the building fire staff, irrespective of the number of remote stations installed. Obtain the approval of the SO for the number of keys to be provided.
 - (iv) Provide remote handset stations finished in red and labelled 'EMERGENCY FIRE TELEPHONE'.
- (c) Remote Handset Jack Stations and Portable Handsets
 - (i) Provide remote jack stations suitable for mounting onto a standard single gang electrical box.
 - (ii) Provide portable handsets constructed of red high impact ABS or other suitable plastic material complete with retractable cord fitted with a plug to suit the jack station.
 - (iii) Provide an appropriate number of portable handsets for the building fire staff, irrespective of the number of remote jack stations installed. Obtain the approval of the SO for the number of portable handsets to be provided.
 - (iv) Provide remote handset jack stations finished in red and labelled 'EMERGENCY FIRE TELEPHONE' or indicated by a telephone symbol.
- (d) System Wiring
 - (i) Make all interconnections to the remote stations on a star basis, with a separate pair of cables running to each remote station.
 - (ii) Ensure that all wiring is fire rated for a minimum of 1 hour to SS 299 in accordance with SS CP 25.

G7.18.5 System Operation

- (a) Calling a Remote from the Master Station
 - (i) When the zone select button for the remote station is pressed, the LED indicator will flash. When the remote station answers, the flashing LED will change to a steady light. This may be repeated for conference calls to multiple stations. To terminate the connection(s) both the master and remote handset(s) have to be replaced on the rest.
- (b) Calling in to the Master from a Remote Station
 - (i) There are two scenarios according to whether the operator at the master station has selected the privacy facility. Basically the privacy facility determines whether multiple incoming calls are automatically connected into a conference call or manually by the operator at the master station.
 - (ii) Any remote station calling in to the master station will cause the respective zone LED to start to flash as well as sounding a distinct audible signal at the control console. When the master station picks up the handset, a connection will be established and the zone indicator will change to a steady light and silence the audible alarm.
 - (iii) If the privacy facility is not selected and a second remote station calls in, it will automatically join in the conversation. The zone indicator LEDs will be steady while the connections are maintained. The scenario will be repeated for further incoming calls.
 - (iv) If the privacy facility is selected, a second remote station calling in to the master station will hear the busy tone and will not get connected automatically. The respective zone LED will start to flash to indicate that there is another incoming call. Pressing the zone select button will join this second remote station into a conference call and the zone indicator LEDs will be steady while the connections are maintained. The scenario will be repeated for further incoming calls.
- (c) Voice Recording Function
 - (i) Whenever a fire alarm signal is initiated the recorder will automatically start recording all signals and broadcast messages and from the PA system on one channel and all voice traffic over the intercom system on the other. All signal inputs and recording status are displayed on the interface unit at the master station.

G7.18.6 Testing & Commissioning

- (a) Conduct the following tests after installation on site, witnessed by the S.O./Engineer or his representative and submitted to the S.O..
 - (i) Check the handsets by transmitting calls between the remote handsets throughout the building and the master handset on both private and conference calls.
 - (ii) Check the recording system.

END OF SECTION